Microsoft Sentinel SOC Lab

Windows Security Monitoring, Detection Engineering & Incident Validation

Cloud-based SIEM Implementation using Microsoft Azure

1. Project Overview

This project demonstrates the end-to-end deployment of a cloud-based Security Information and Event Management (SIEM) solution using Microsoft Sentinel, focused on real-world Security Operations Centre (SOC) workflows.

A local Windows endpoint was onboarded to Azure via Azure Arc and integrated with Microsoft Sentinel via the Azure Monitor Agent (AMA). The lab enables real-time ingestion, analysis, and detection of Windows Security Events, followed by validation through simulated attack scenarios.

The objective of this project is to replicate the practical responsibilities of SOC analysts, including log collection, event analysis, detection rule creation, and incident validation.

2. Objectives

- Deploy and configure Microsoft Sentinel as a cloud SIEM platform
- Onboard a local Windows endpoint using Azure Arc
- Enable ingestion of Windows Security Event logs via Azure Monitor Agent
- Perform log analysis using Kusto Query Language (KQL)
- Design and implement detection rules based on real attack scenarios
- Simulate attacks and validate end-to-end detection capability
- Investigate alerts and incidents within the Sentinel SOC dashboard

3. Architecture

3.1 Components Used

Component	Technology
SIEM Platform	Microsoft Sentinel
Log Storage	Log Analytics Workspace
Endpoint Integration	Azure Arc (Hybrid Machine)

Agent	Azure Monitor Agent (AMA)
Data Source	Windows Security Event Logs

3.2 Data Flow

The following pipeline describes how security events are collected, processed, and surfaced as incidents:

Local Windows Machine → Azure Arc → Azure Monitor Agent → Data Collection Rule → Log Analytics Workspace → Microsoft Sentinel → Analytics Rules → Incidents

4. Environment Setup

4.1 Microsoft Sentinel Deployment

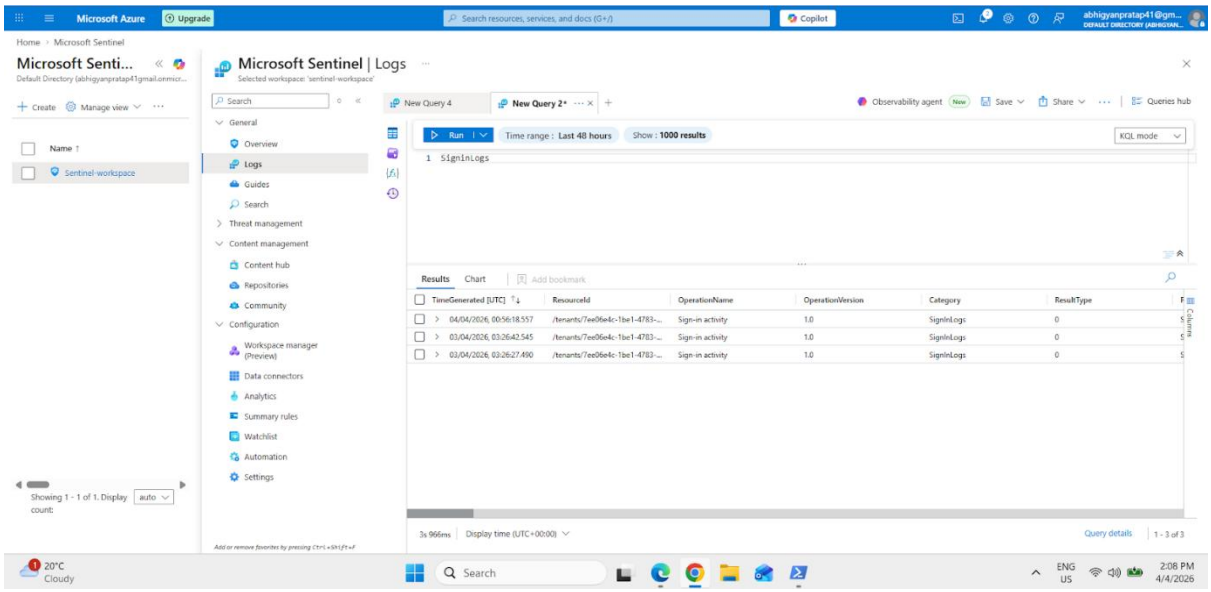
- Created a Log Analytics Workspace in Azure
- Enabled Microsoft Sentinel on the workspace
- Configured Data Connectors for Windows Security Events

4.2 Azure Arc Onboarding

The local Windows machine was registered as a hybrid Azure resource using Azure Arc. The Azure Connected Machine Agent was installed via a PowerShell script executed with administrator privileges.

- Downloaded and executed the Azure Arc onboarding script via PowerShell
- Authenticated via browser pop-up during agent installation
- Verified successful connection in the Azure Arc portal

Screenshot: Azure Sentinel Logs Interface & Initial Log Query



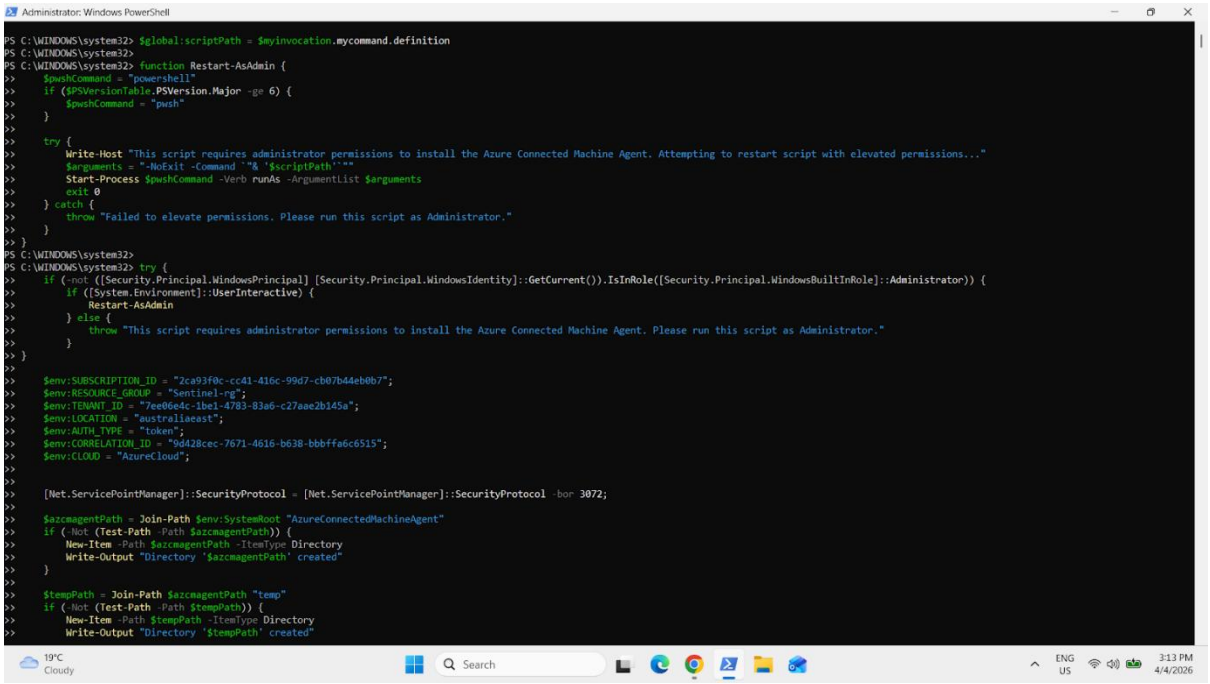
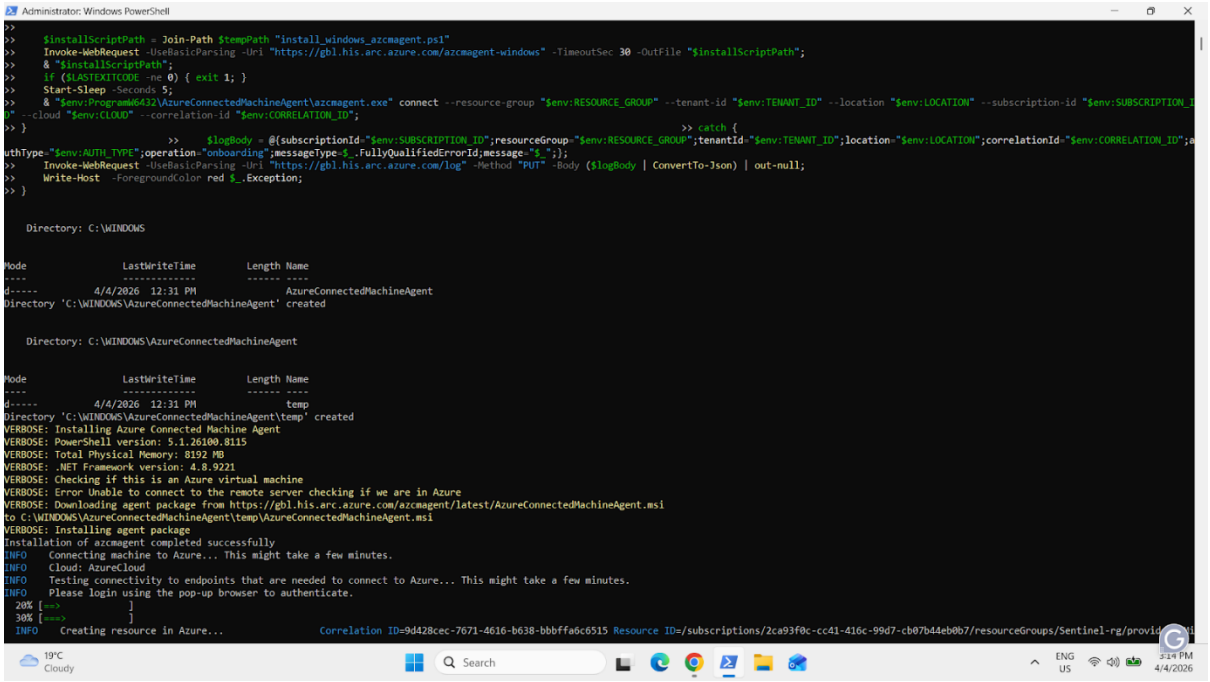


Fig. 1 — Microsoft Sentinel Logs interface showing Sign-inLogs query results, and Azure Arc PowerShell onboarding script execution.

Screenshot: Azure Arc Agent Installation & Connection



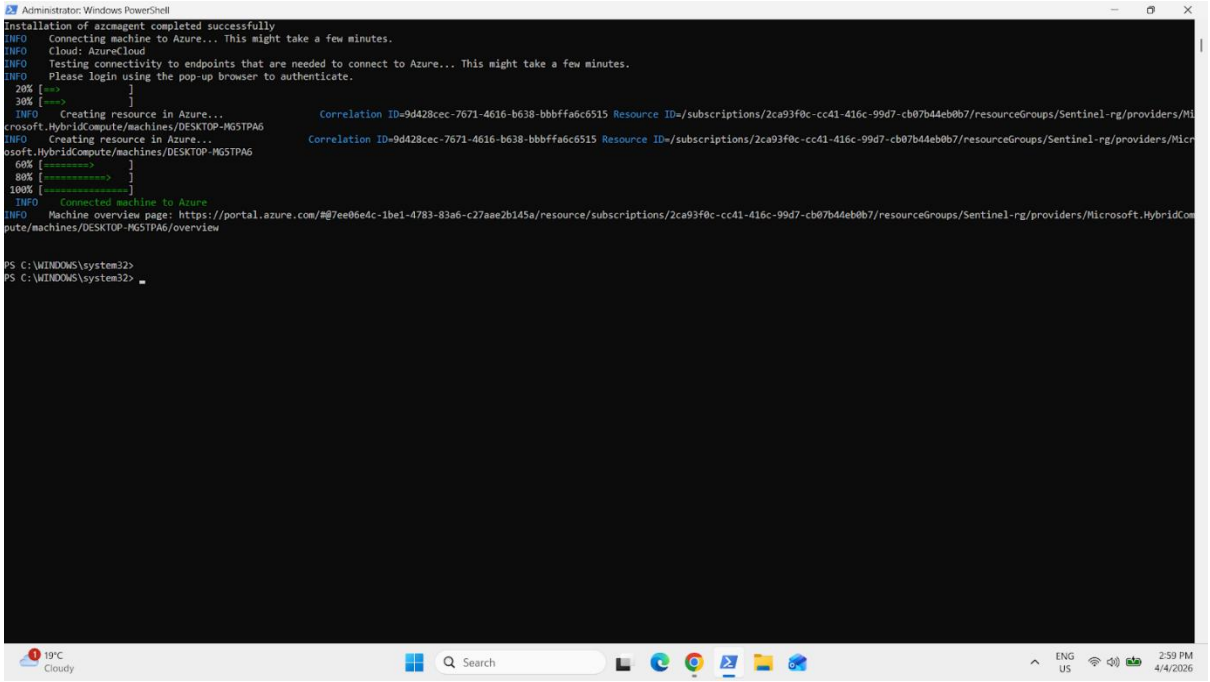


Fig. 2 — PowerShell output confirming Azure Connected Machine Agent installation, directory creation, and successful connection to Azure Arc.

4.3 Data Collection Configuration

- Created a Data Collection Rule (DCR) in Azure Monitor
- Selected Windows Security Events (All Events) as the data source
- Linked the Azure Arc machine to the DCR
- Verified log ingestion in Log Analytics Workspace

5. Data Ingestion & Log Sources

Windows Security logs were collected in real time using the Azure Monitor Agent (AMA) and routed through the Data Collection Rule into the Log Analytics Workspace. Key Event IDs were monitored to cover the most critical threat vectors.

Event ID	Description	Use Case
4624	Successful Logon	User authentication tracking
4625	Failed Logon	Brute-force detection
4672	Special Privileges Assigned	Privileged access monitoring
4688	Process Creation	Suspicious process execution
4732	User Added to Group	Privilege escalation detection

6. Log Analysis Using KQL

Kusto Query Language (KQL) was used to query, filter, and analyse ingested security logs within the Log Analytics Workspace.

6.1 Authentication Analysis

Queries authentication events to identify both successful and failed logons:

```
SecurityEvent
| where EventID == 4624 or EventID == 4625
```

6.2 Privilege & Process Monitoring

Queries for special privilege assignments and new process creation events:

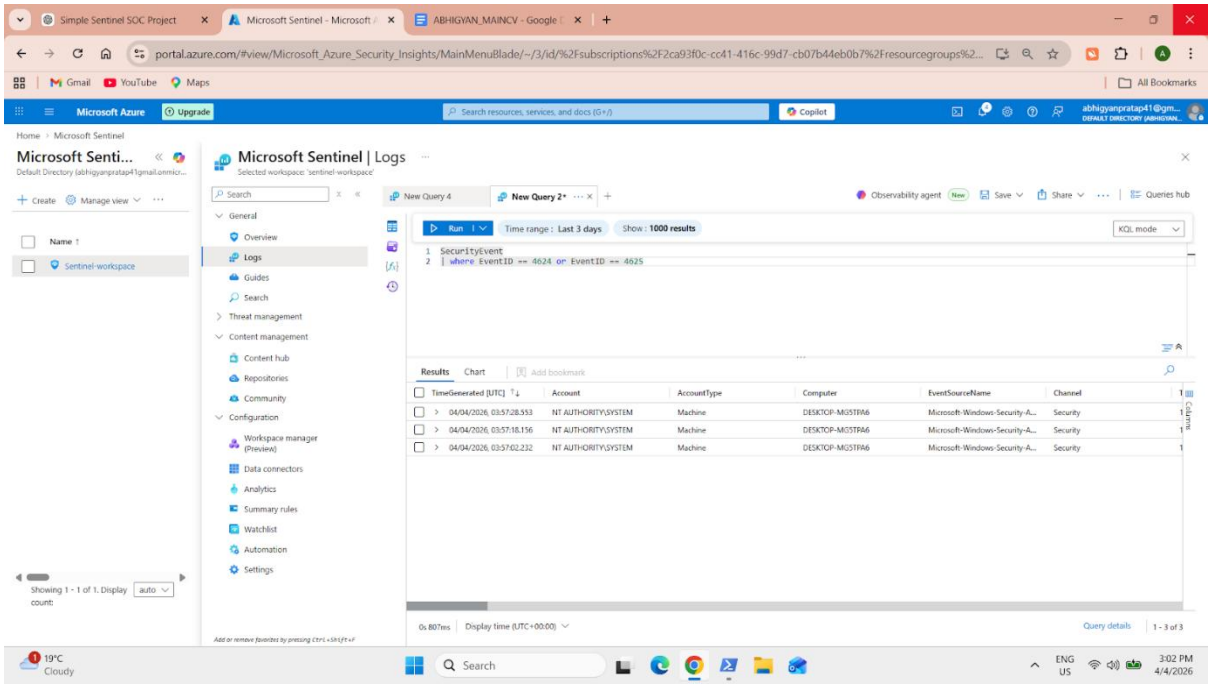
```
SecurityEvent
| where EventID in (4672, 4688)
```

6.3 Aggregation of Failed Logins

Aggregates failed login counts per account in 5-minute time windows — the basis for brute-force detection:

```
SecurityEvent
| where EventID == 4625
| summarize FailedAttempts=count() by Account, bin(TimeGenerated, 5m)
```

Screenshot: KQL Queries in Log Analytics



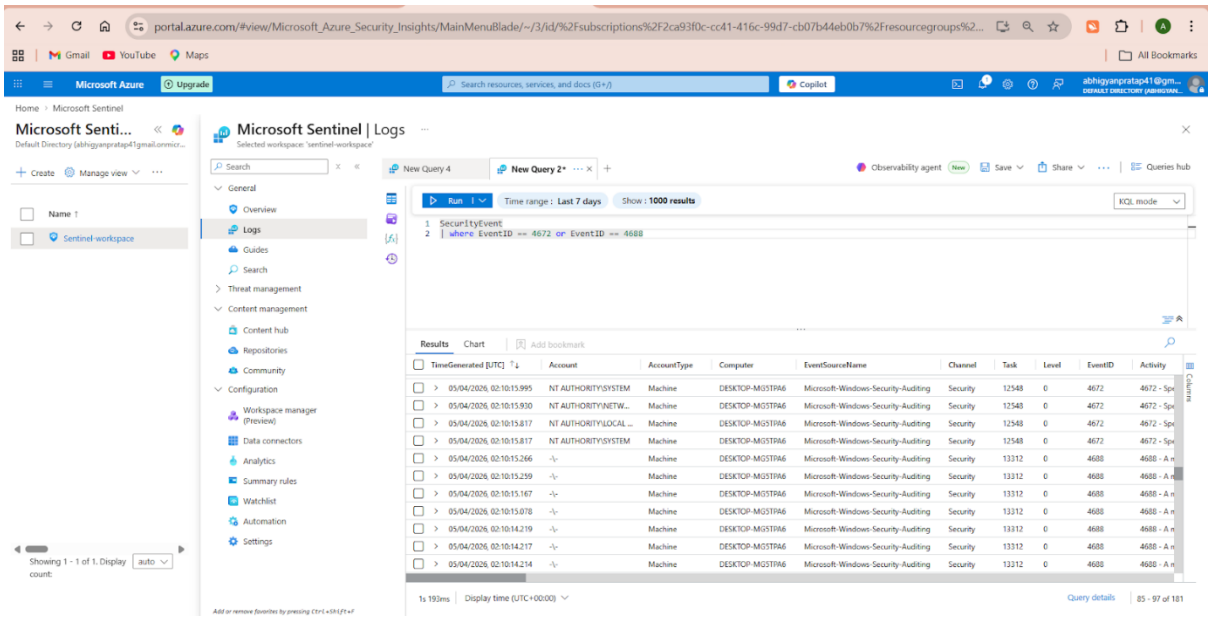


Fig. 3 — KQL queries running against SecurityEvent logs, showing Event ID 4624/4625 authentication events and Event ID 4672/4688 privilege/process events across the DESKTOP-MGSTPA6 host.

7. Detection Engineering

Two high-impact detection rules were created in Microsoft Sentinel Analytics to simulate real SOC use cases. Both rules are configured with High severity and are enabled as active analytics rules.

7.1 Rule 1 — Brute Force Login Detection

This rule detects multiple failed login attempts targeting the same account within a short time window, a pattern consistent with password guessing and brute-force attacks.

Rule Logic (KQL)

```
SecurityEvent
| where EventID == 4625
| summarize FailedAttempts = count() by Account, Computer, bin(TimeGenerated, 5m)
| where FailedAttempts > 5
```

Configuration

- Severity: High
- Run Frequency: Every 5 minutes
- Lookup Period: Last 5 minutes
- Trigger Condition: More than 5 failed logins in the window
- Tactic: Credential Access

Purpose

Identifies potential password guessing or brute-force attacks targeting user accounts on monitored endpoints.

7.2 Rule 2 — Privilege Escalation Detection

This rule detects when any user account is added to the local Administrators group, which is a common indicator of privilege escalation and attacker persistence activity.

Rule Logic (KQL)

```
SecurityEvent
| where EventID == 4732
| where TargetUserName contains "Administrators"
```

Configuration

- Severity: High
- Trigger Condition: Any match (zero-threshold — every occurrence triggers an alert)
- Tactic: Privilege Escalation

Purpose

Identifies unauthorised privilege escalation attempts or attacker persistence mechanisms where a new account is promoted to administrator.

8. Attack Simulation & Validation

Both detection rules were validated by performing controlled attack simulations on the monitored Windows endpoint.

8.1 Brute Force Attack Simulation

The Windows system was locked using Win + L, and incorrect passwords were entered multiple times within a short window to generate Event ID 4625 entries at a sufficient rate to trigger the detection rule.

Result

- Multiple Event ID 4625 entries generated and ingested into Log Analytics
- Brute Force detection rule triggered successfully
- Alert generated in Microsoft Sentinel with High severity
- Incident created and visible in the SOC Incidents dashboard

8.2 Privilege Escalation Simulation

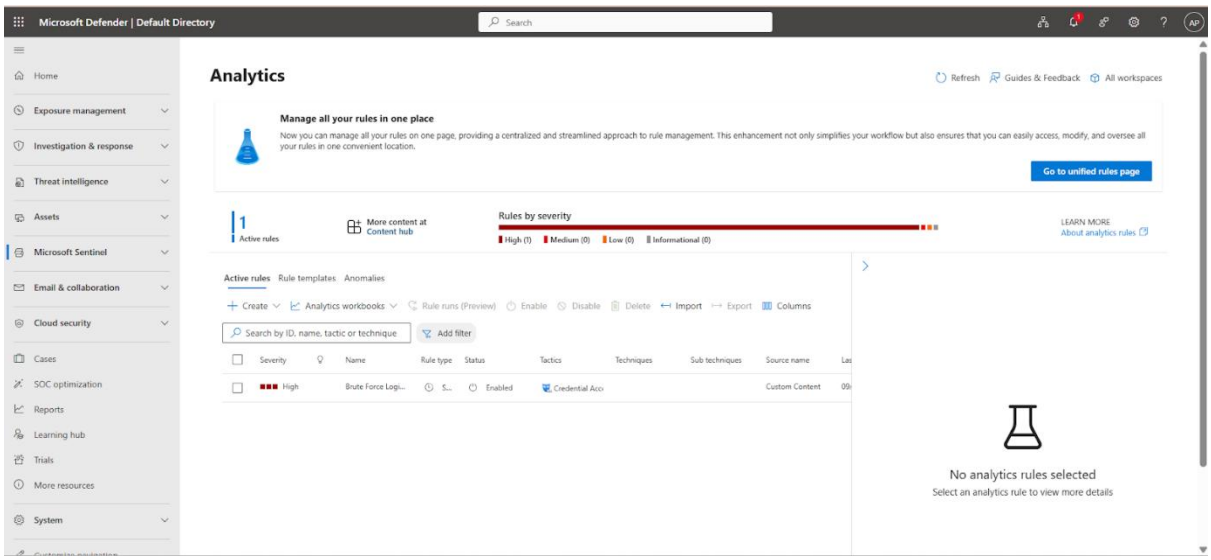
A new local user account was created and immediately added to the local Administrators group via PowerShell commands:

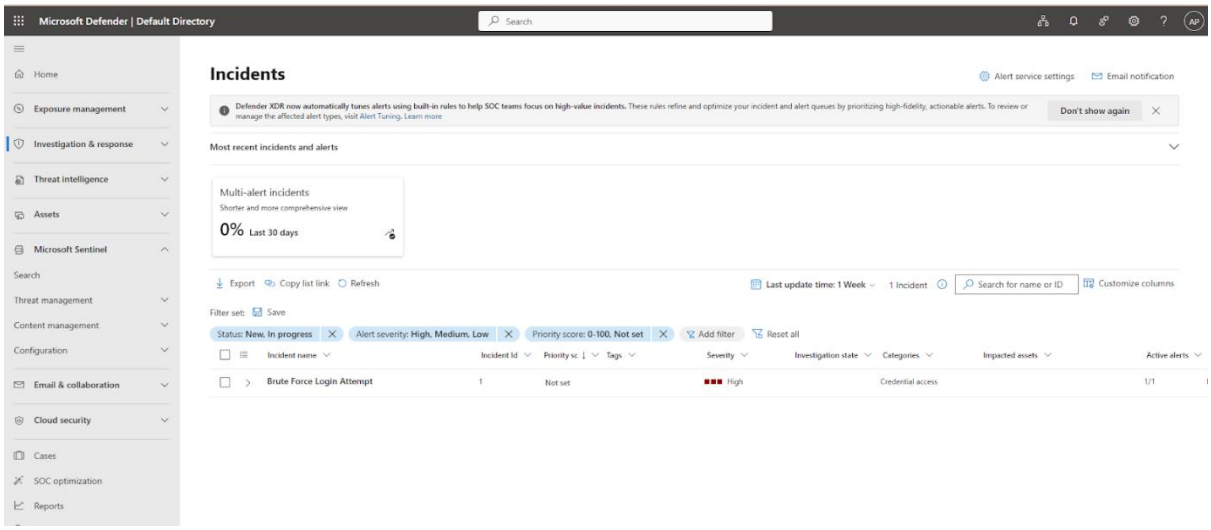
```
net user sentineltest Password123! /add
net localgroup administrators sentineltest /add
```

Result

- Event ID 4732 generated and ingested into Log Analytics
- Privilege Escalation rule triggered immediately
- High-severity alert and incident created in Sentinel

Screenshot: Brute Force — KQL Verification & Active Analytics Rules





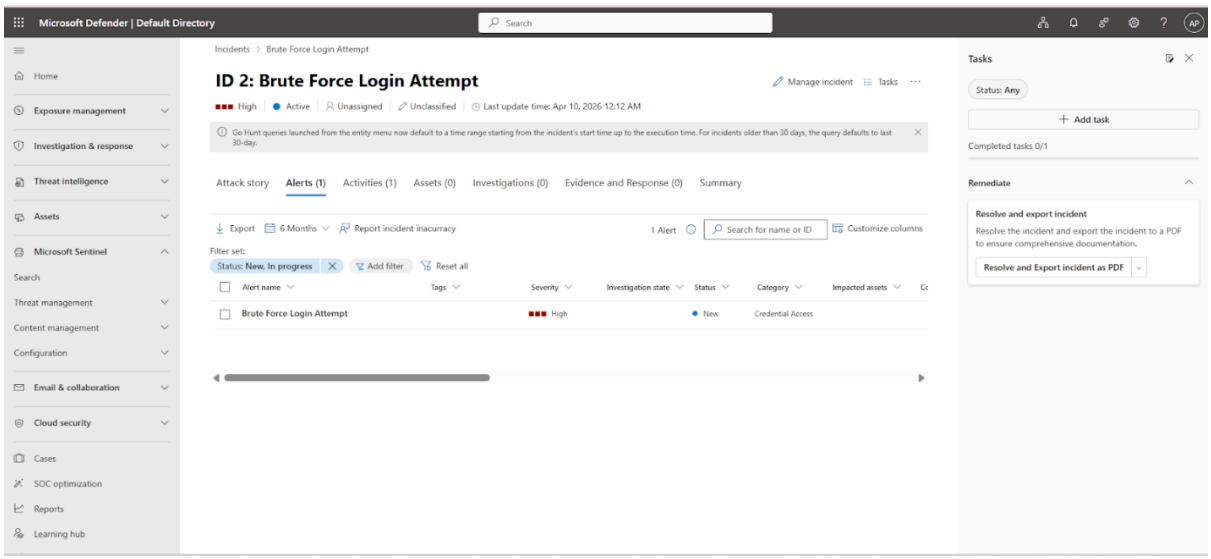
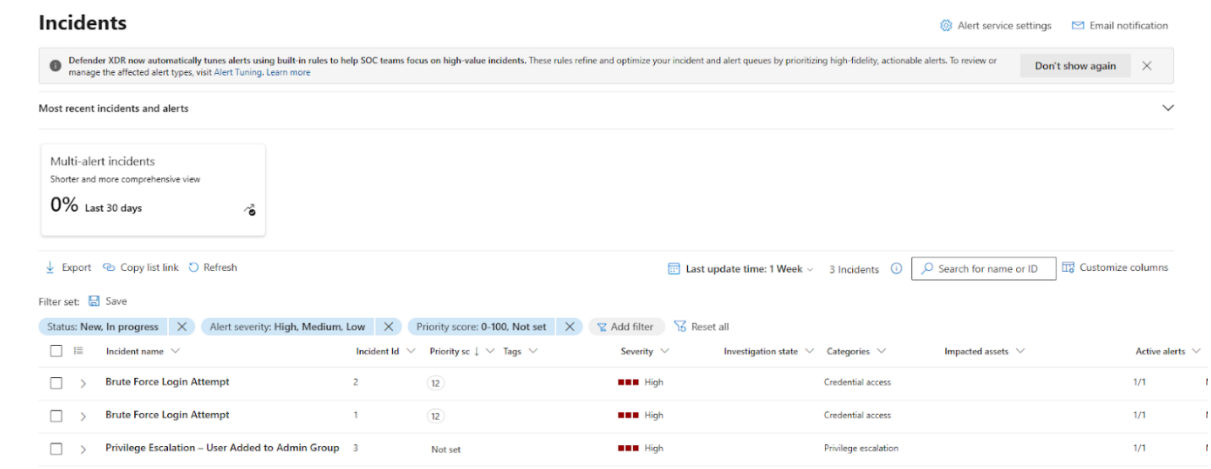


Fig. 4 — Active Analytics rules panel showing the Brute Force Login rule enabled, Incidents dashboard displaying the triggered incident, and the detailed incident view for "ID 2: Brute Force Login Attempt" with High severity.

Screenshot: Privilege Escalation — Incidents & Alert Detail



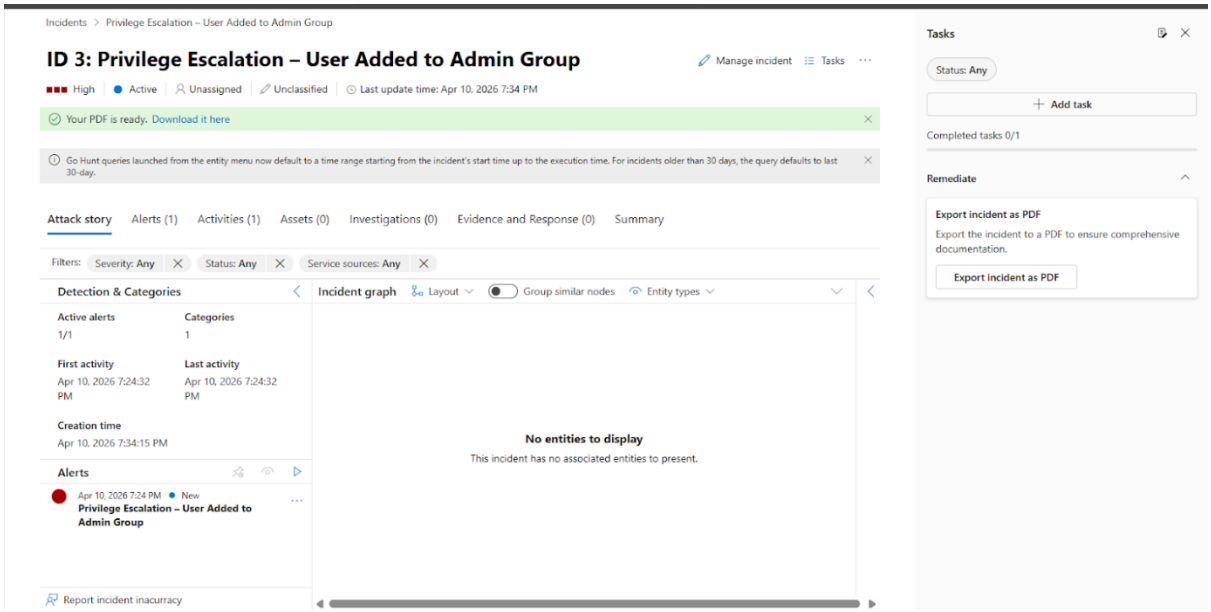
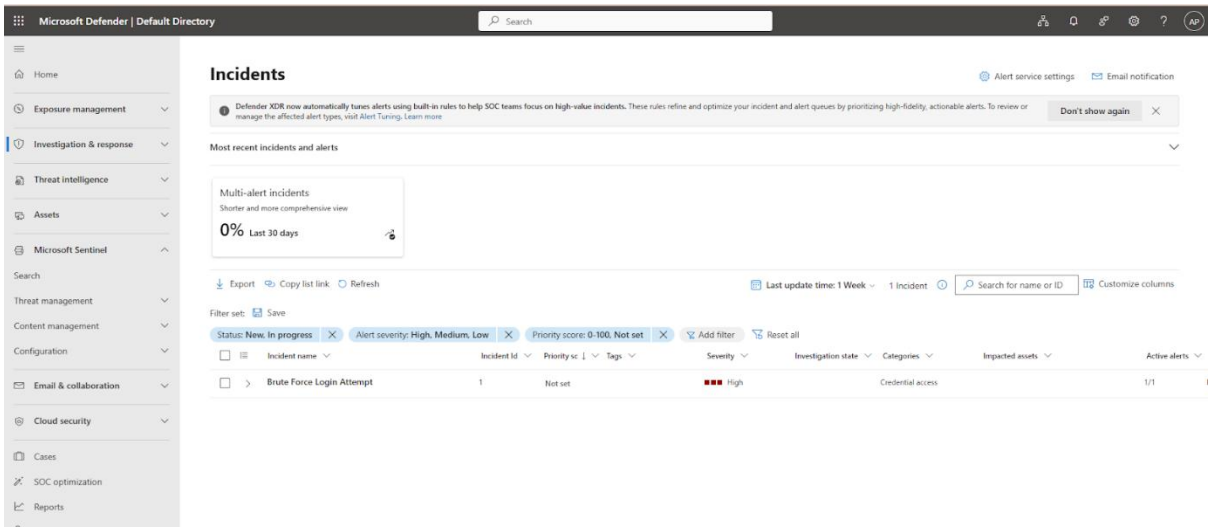
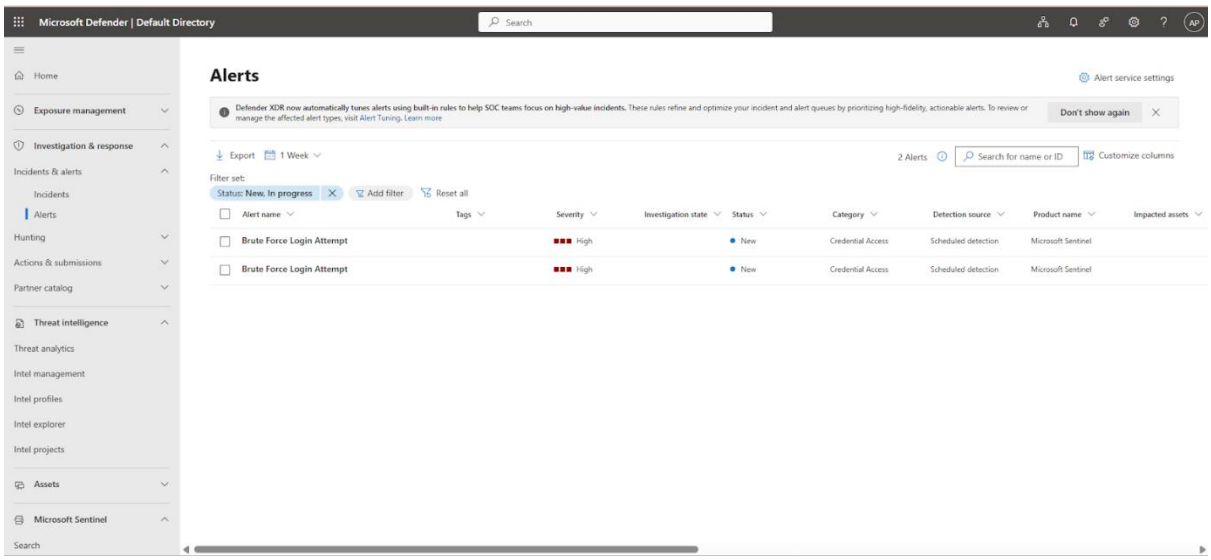


Fig. 5 — Sentinel Incidents panel showing all three incidents, a detailed view of "ID 3: Privilege Escalation — User Added to Admin Group," and the associated alert with activity timeline.

Screenshot: Alert Views & Rule Metadata





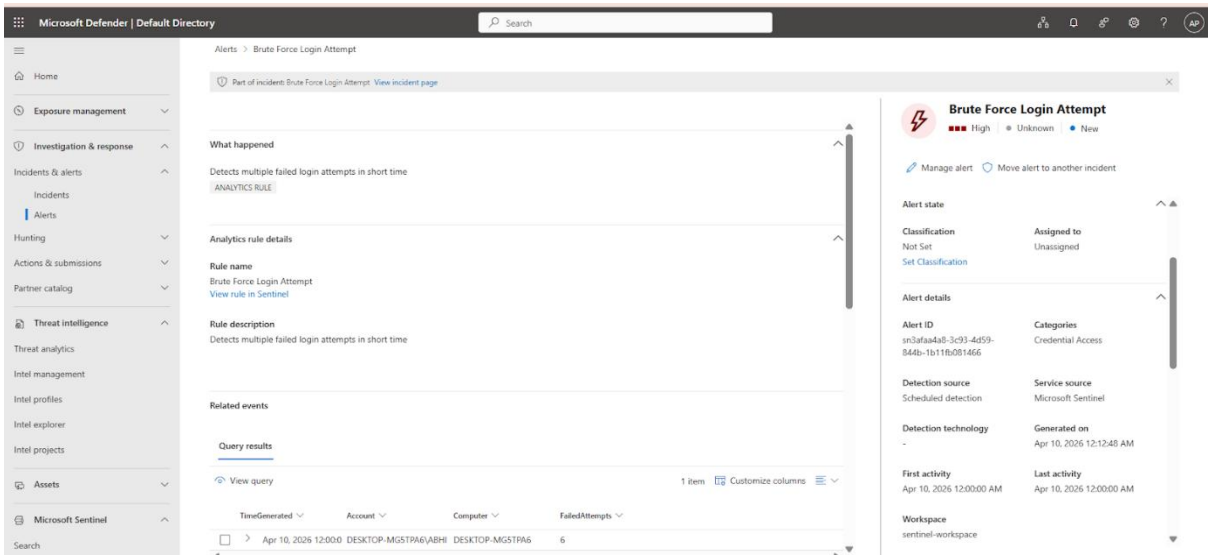
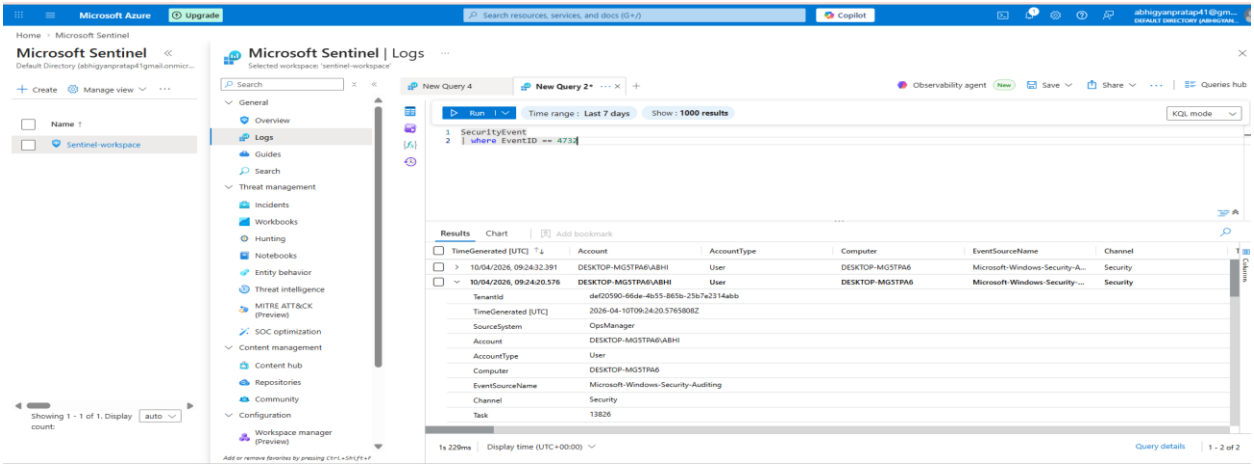
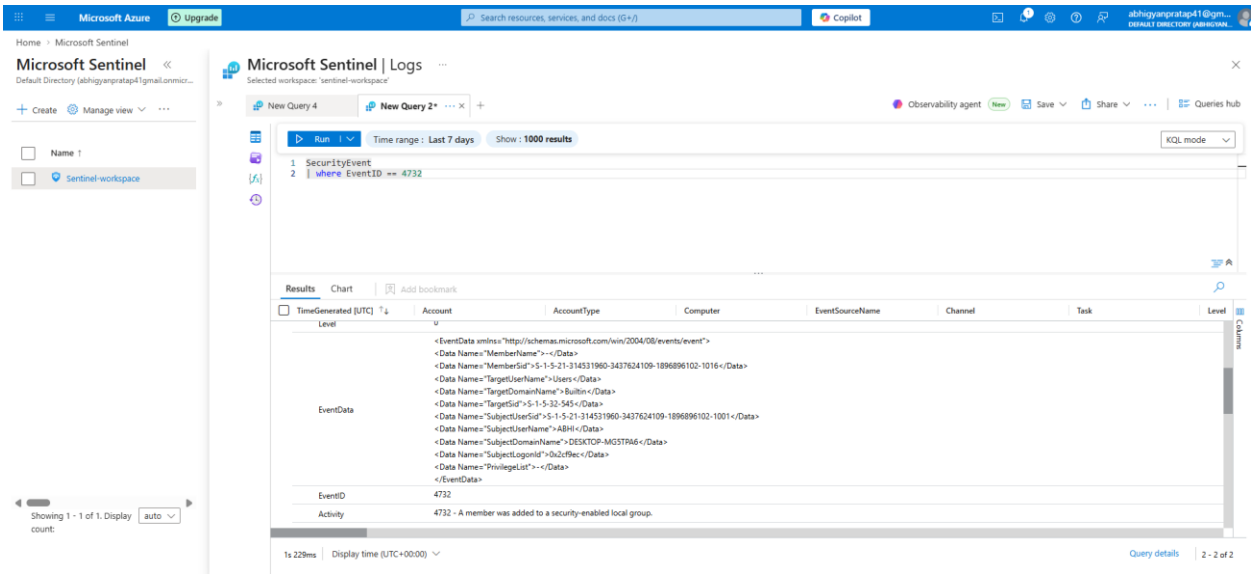
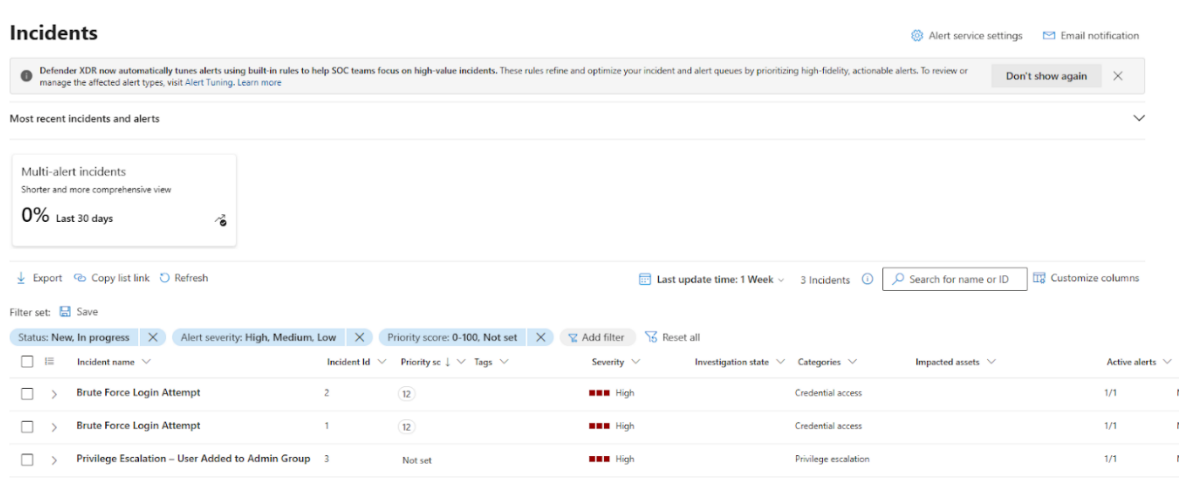


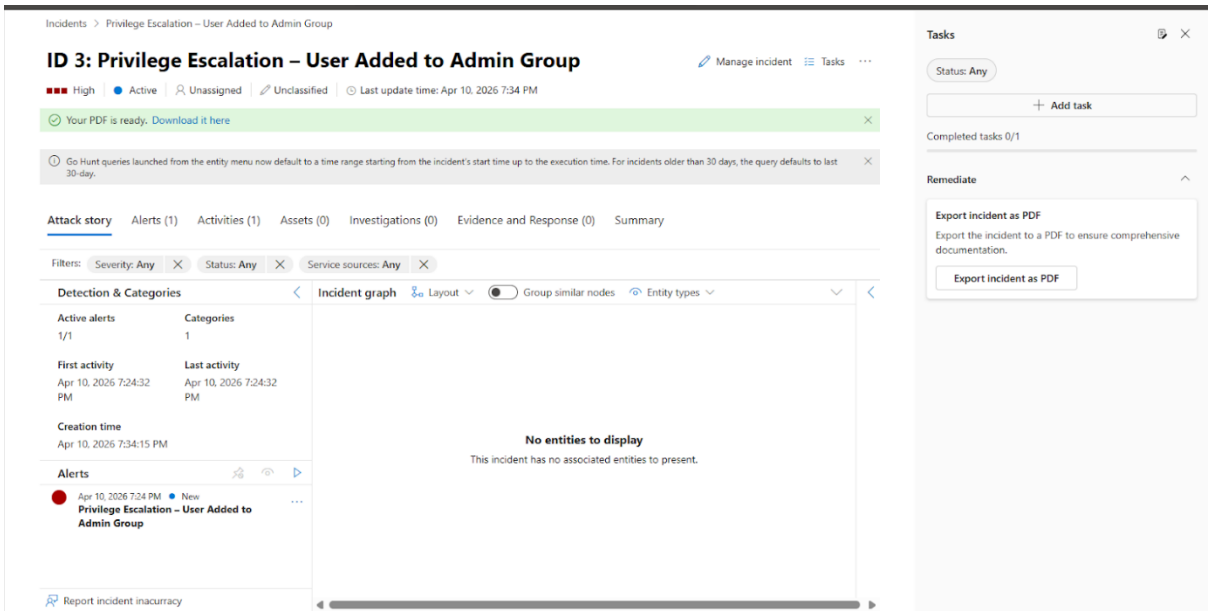
Fig. 6 — Alerts panel listing Brute Force Login Attempt detections, alert detail panel confirming the analytics rule triggered, and incident detail showing six failed login attempts aggregated within the 5-minute window.

Screenshot: Privilege Escalation — Alert Evidence & Event Data









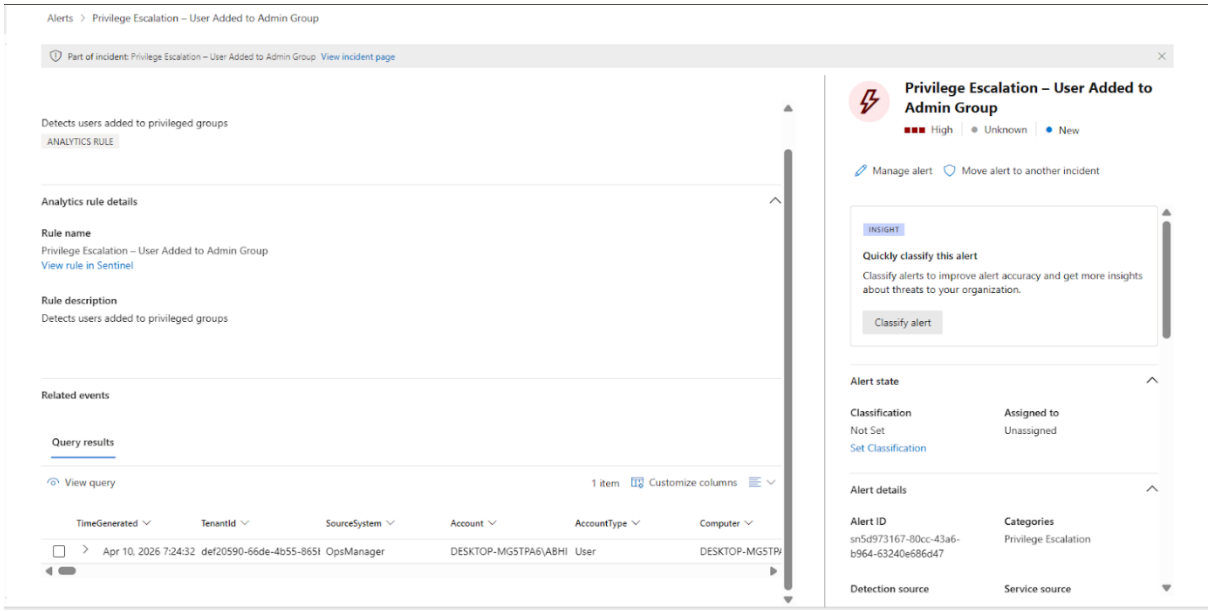


Fig. 7 — Alert detail for "Privilege Escalation — User Added to Admin Group" showing the analytics rule metadata, query results, and Event ID 4732 evidence with the impacted computer (DESKTOP-MGSTPA6) and affected account.

Screenshot: PowerShell — Privilege Escalation Commands

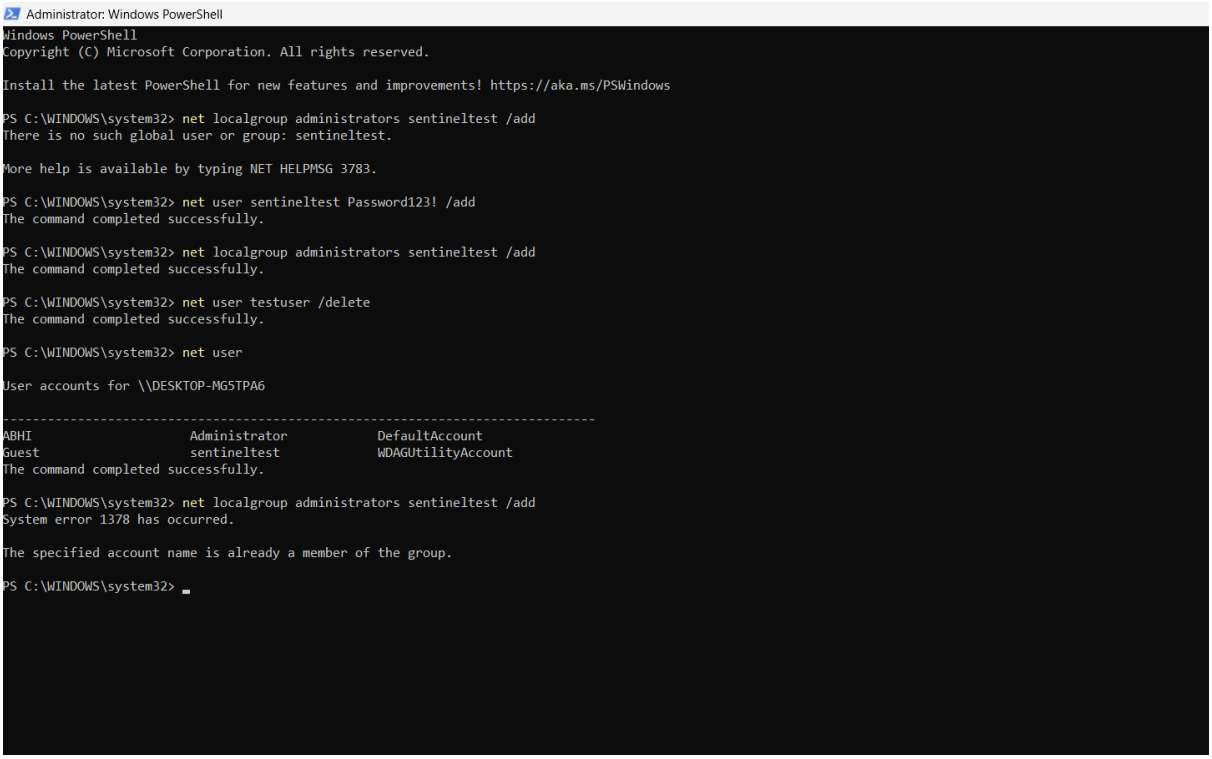


Fig. 8 — PowerShell terminal confirming execution of net user and net localgroup commands; user "sentineltest" successfully created and added to Administrators group on DESKTOP-MGSTPA6.

9. Incident Investigation

All alerts generated by the detection rules were investigated within the Microsoft Sentinel Incidents dashboard. The following observations were made during the investigation process.

- Alerts correlated correctly with the events generated during attack simulations
- Time-based aggregation in the brute-force rule confirmed 6 failed attempts within a single 5-minute window
- Incident filtering initially hid valid alerts due to default priority scoring — adjusting filter settings (Status: New, In Progress; Severity: High, Medium, Low) revealed the correct detections
- Incident "ID 3: Privilege Escalation — User Added to Admin Group" showed the affected account, source computer, and Event ID 4732 evidence in the query results panel
- Alert classification was set to New and remained Unassigned — in a production SOC, this would be triaged and assigned to an analyst

10. Key Learnings

- End-to-end SIEM pipeline implementation in a cloud environment using Azure-native tooling
- Practical experience with Azure Arc for onboarding hybrid (non-Azure) endpoints into a cloud monitoring solution

- Deep understanding of Windows Security Event logging, key Event IDs, and their relevance to threat detection
- Hands-on detection engineering using KQL, translating threat scenarios into working analytics rules
- Understanding of how alert aggregation windows affect rule sensitivity and specificity
- Practical exposure to SOC incident management workflows: alert triage, incident review, and evidence analysis

11. Conclusion

This project successfully demonstrates real-world SOC analyst capabilities by implementing a fully functional SIEM pipeline using Microsoft Sentinel. From onboarding a local Windows endpoint and ingesting security event logs to designing custom detection rules and validating incidents through controlled attack simulations, the project reflects practical security monitoring and threat detection workflows used in modern SOC environments.

The ability to detect brute-force login attempts and privilege escalation activity through purpose-built KQL analytics rules highlights the effectiveness of the implemented detections. Both rules fired correctly against simulated attack scenarios and produced actionable incidents with supporting evidence within the Sentinel dashboard.

This project demonstrates a strong foundation in detection engineering, Windows event log analysis, cloud SIEM architecture, and incident investigation — core competencies for a practising SOC analyst.